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Daenet Semantic Testing Framework

Antonio Leone

Abstract

Large Language Models (LLMs) are increasingly integrated into modern software systems, creating new challenges for quality assurance and automated testing. Traditional unit testing relies on deterministic outputs and equality-based assertions, which are insufficient when validating natural-language responses. This paper introduces the Daenet Semantic Testing Framework, a methodology that combines embedding-based similarity metrics with reasoning-based evaluation to assess semantic equivalence between expected and generated outputs.

Introduction

Software development relies heavily on testing to ensure correctness, reliability, and maintainability.

Conventional unit tests compare an actual result with an expected result using deterministic assertions. While this approach is effective for numerical values and structured outputs, it becomes problematic when testing systems powered by LLMs. Two responses may express the same meaning using different wording, causing traditional equality assertions to fail despite semantic equivalence.

Semantic Similarity

The first challenge is transforming natural language into a numerical representation that preserves semantic information. This is achieved using embedding models, which map text into high-dimensional vectors. Both the expected and actual responses are converted into embeddings. Similarity can then be measured using cosine similarity. A score close to 1 indicates strong semantic alignment, a score near 0 indicates little similarity, and negative values suggest opposing meanings. This enables automated comparison of natural-language responses beyond exact string matching.

Semantic Meaning

Semantic similarity alone is insufficient. Consider the statements “A man bites a dog” and “A dog bites a man.” Although the wording is highly similar, the meanings are fundamentally different. Therefore, semantic evaluation must also consider reasoning and context. The Daenet Semantic Testing Framework introduces an LLM-as-a-Judge approach. The expected output, actual output, and an evaluation rubric are provided to a language model. The model assesses whether the generated response satisfies the semantic requirements defined by the rubric and returns a structured score ranging from 0.0 to 1.0.

Evaluation Framework

A predefined prompt ensures consistent and machine-readable evaluation. The framework requests concise reasoning and a standardized score, enabling automatic parsing and integration into test pipelines. This approach combines quantitative similarity metrics with qualitative reasoning-based assessment, resulting in a more reliable evaluation process for AI-generated content.

Research Perspective

The framework is also the subject of ongoing academic research. Future work includes benchmarking different LLMs as judge models, evaluating their reliability, consistency, and trustworthiness. Additional research investigates novel metrics that may extend current evaluation methods and provide deeper insights into LLM performance and semantic correctness.

Conclusion

The Daenet Semantic Testing Framework addresses a critical challenge in AI-enabled software testing: validating semantic equivalence rather than literal equality. By combining vector-based similarity measurements with reasoning-driven LLM evaluation, the framework provides a robust methodology for testing modern AI applications.